

IA Lect 5

Subject: Install Brim

[Brimdata](#) has introduced the Brim security tool, an open-source **desktop application** that can be used to analyze structured network traffic. The Brim tool uses Zeek logs to examine the traffic. The *Brim* tool now has been renamed to *Zui*

Your task is to install Brim on an Ubuntu VM along with Zeek. You will need to:

- Install Ubuntu 22.04
- Install Wireshark (optional but highly recommended)
- Install Zeek
- Install Brim.

Deliverable: Provide the list of all commands used along with screenshots. Live Demonstration using the provided pcap file.

Help:

<https://kifarunix.com/analyze-network-traffic-using-brim-security/>

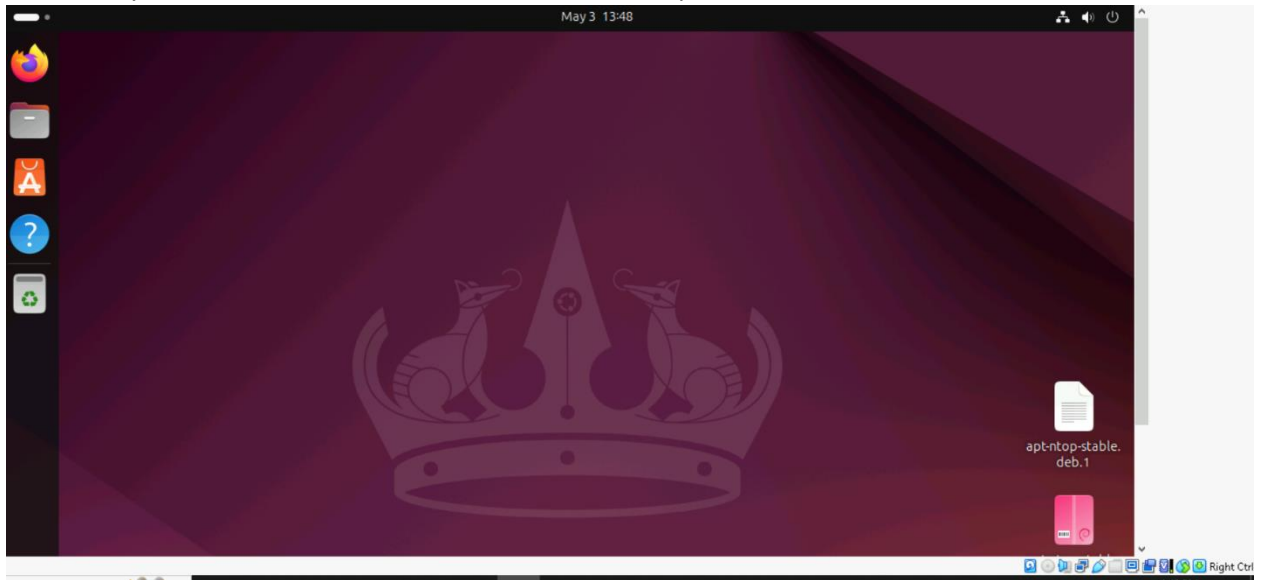
<https://kifarunix.com/install-zeek-on-ubuntu/>

Delivery

Create a **pdf** file and name it IALect5_xxxxxxxxx where xxxxxxxxx is your name

PS. Provide screenshots.

- 1) I've already installed the Ubuntu Machine before, in our previous IA



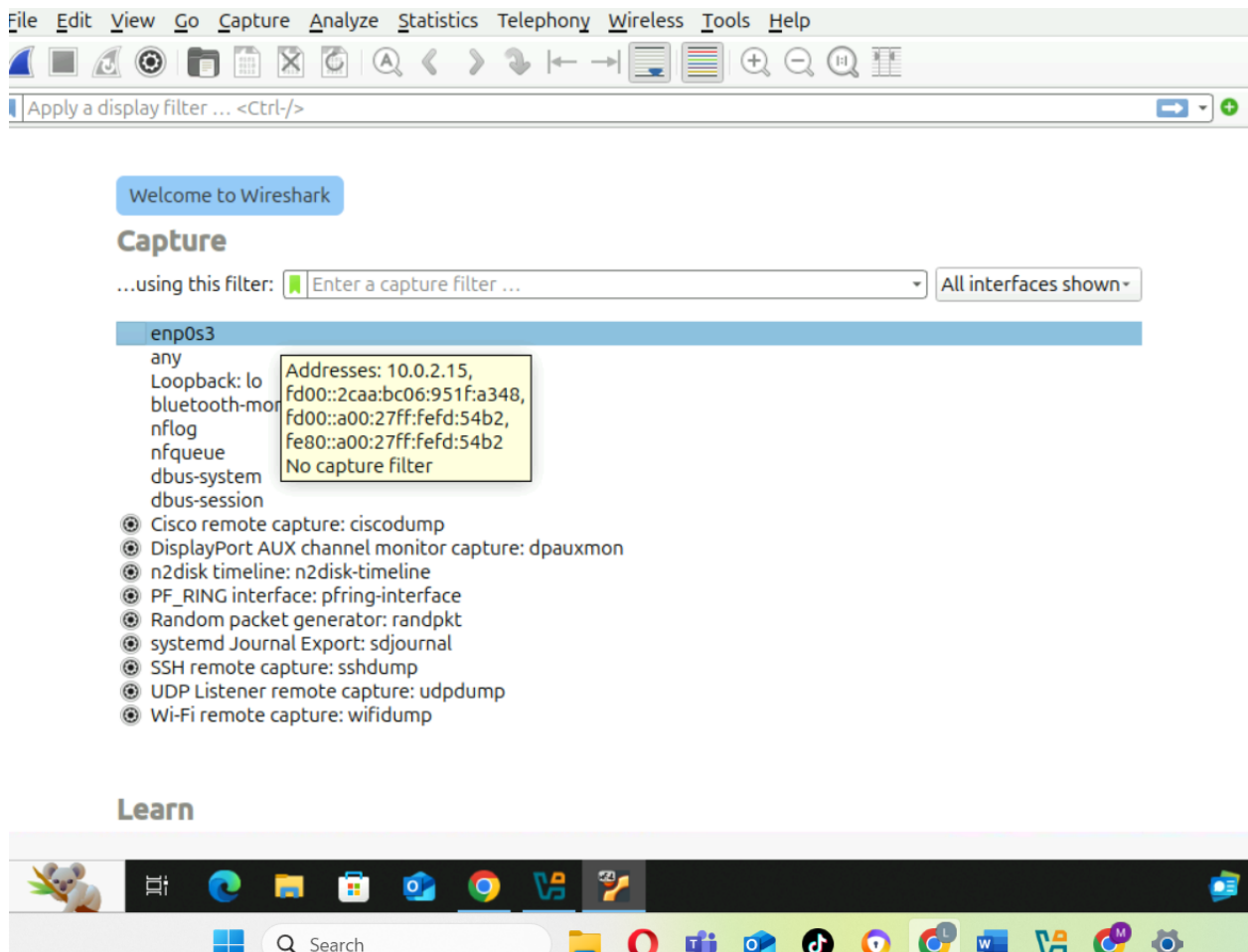
- 2) Wireshark

I've updated the system with the `sudo apt update && sudo apt upgrade -y`

```
vboxuser@UbuntuLTS: ~/Desktop
vboxuser@UbuntuLTS:~/Desktop$ sudo apt update && sudo apt upgrade -y
Hit:1 http://gr.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://gr.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:4 http://gr.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Hit:5 https://packages.ntop.org/apt-stable/24.04 x64/ InRelease [2,153 B]
Hit:6 https://packages.ntop.org/apt-stable/24.04 all/ InRelease [2,153 B]
Hit:7 http://gr.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,057 kB]
Hit:8 https://packages.ntop.org/apt-stable/24.04 x64/ Packages [3,937 B]
Hit:9 https://packages.ntop.org/apt-stable/24.04 all/ Packages [2,808 B]
Hit:10 http://gr.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [227 kB]
Hit:11 http://gr.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [161 kB]
Hit:12 http://gr.archive.ubuntu.com/ubuntu noble-updates/main Icons (48x48) [34.7 kB]
Hit:13 http://gr.archive.ubuntu.com/ubuntu noble-updates/main Icons (64x64) [49.6 kB]
Hit:14 http://gr.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1,062 kB]
Hit:15 http://gr.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [220 kB]
Hit:16 http://gr.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Hit:17 http://gr.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,060 kB]
Hit:18 http://gr.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [268 kB]
Hit:19 http://gr.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [377 kB]
Hit:20 http://gr.archive.ubuntu.com/ubuntu noble-updates/universe Icons (48x48) [226 kB]
Hit:21 http://gr.archive.ubuntu.com/ubuntu noble-updates/universe Icons (64x64) [350 kB]
Hit:22 http://gr.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [21.7 kB]
Hit:23 http://gr.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Hit:24 http://gr.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,064 B]
Hit:25 http://gr.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Hit:26 http://gr.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [27.1 kB]
```

I've installed Wireshark with the command `sudo apt install wireshark -y`

```
vboxuser@UbuntuLTS:~/Desktop$ sudo apt install wireshark -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-57 linux-headers-6.8.0-57-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  i965-va-driver intel-media-va-driver libaacs0 libavcodec60 libavformat60 libavutil58 libb2-1 libbcg729-0 libbdplus0
  libbluray2 libchromaprint1 libcjson1 libcodec2-1.2 libdav1d7 libdouble-conversion3 libgme0 libgsm1 libhwy1t64
  libigdgmm12 libjxl0.7 liblua5.2-0 libmbedcrypto7t64 libmd4c0 libminizip1t64 libnghttp3-3 libopencore-amrnb0
  libopenmpt0t64 libpcre2-16-0 libqt6core5compat6 libqt6core6t64 libqt6dbus6t64 libqt6gui6t64 libqt6multimedia6
  libqt6network6t64 libqt6opengl6t64 libqt6printsupport6t64 libqt6qml6 libqt6qmlmodels6 libqt6quick6 libqt6svg6
  libqt6waylandclient6 libqt6waylandcompositor6 libqt6waylandeglclienthwinintegration6
  libqt6waylandeglcompositorhwinintegration6 libqt6widgets6t64 libqt6wlshellintegration6 librabbitmq4 librav1e0 librist4
  libshine3 libsmi2t64 libsnappy1v5 libsoxr0 libspandsp2t64 libsrtp1.5-gnutls libssh-gcrypt-4 libsvtav1enc1d1
  libswresample4 libswscale7 libts0t64 libudfread0 libva-drm2 libva-x11-2 libva2 libvdpau1 libvpl2 libwireshark-data
  libwireshark17t64 libwiretap14t64 libwsutil15t64 libx264-164 libx265-199 libxvidcore4 libzvi-common libzvi0t64
```



3) Zeek

Install required dependencies

`sudo apt install cmake make gcc g++ flex bison libpcap-dev libssl-dev python3-dev swig zlib1g-dev -y`

```
vboxuser@UbuntuLTS:~/Desktop$ sudo apt install cmake make gcc g++ bison libpcap-dev libssl-dev python3-dev swig zlib1g-dev -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
make is already the newest version (4.3-4.1build2).
make set to manually installed.
gcc is already the newest version (4:13.2.0-7ubuntu1).
gcc set to manually installed.
g++ is already the newest version (4:13.2.0-7ubuntu1).
g++ set to manually installed.
bison is already the newest version (2:3.8.2+dfsg-1build2).
libpcap-dev is already the newest version (1.10.4-4.1ubuntu3).
libssl-dev is already the newest version (3.0.13-0ubuntu3.5).
zlib1g-dev is already the newest version (1:1.3.dfsg-3.1ubuntu2.1).
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-57 linux-headers-6.8.0-57-generic
```

Download Zeek source `wget https://download.zeek.org/zeek-6.0.0.tar.gz`

```
vboxuser@UbuntuLTS:~/Desktop$ wget https://download.zeek.org/zeek-6.0.0.tar.gz
--2025-05-03 14:06:54-- https://download.zeek.org/zeek-6.0.0.tar.gz
Resolving download.zeek.org (download.zeek.org)... 52.85.223.116, 52.85.223.12, 52.85.223.69, ...
Connecting to download.zeek.org (download.zeek.org)|52.85.223.116|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 60086607 (57M) [application/x-gzip]
Saving to: 'zeek-6.0.0.tar.gz'

zeek-6.0.0.tar.gz          100%[=====>] 57.30M  20.4MB/s   in 2.8s
2025-05-03 14:06:58 (20.4 MB/s) - 'zeek-6.0.0.tar.gz' saved [60086607/60086607]
```

Extract the archive

`tar xzf zeek-6.0.0.tar.gz`

`cd zeek-6.0.0`

Configure, build, and install

`./configure` (This runs the **configure script** in the current directory)

`make -j$(nproc)` (**make** reads the Makefile generated by configure and starts compiling the source code into binaries.

`-j$(nproc)` tells make to use all available CPU cores in parallel.)

`sudo make install`


```

vboxuser@UbuntuLTS:~/Desktop$ tar xzf zeek-6.0.0.tar.gz
vboxuser@UbuntuLTS:~/Desktop$ cd zeek-6.0.0.tar.gz
bash: cd: zeek-6.0.0.tar.gz: Not a directory
vboxuser@UbuntuLTS:~/Desktop$ cd zeek-6.0.0
vboxuser@UbuntuLTS:~/Desktop/zeek-6.0.0$ ./configure
Build Directory : build
Source Directory: /home/vboxuser/Desktop/zeek-6.0.0
Using cmake version 3.28.3

-- The C compiler identification is GNU 13.3.0
-- The CXX compiler identification is GNU 13.3.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Performing Test test_arch_x64
-- Performing Test test_arch_x64 - Success
-- Performing Test test_arch_aarch64
-- Performing Test test_arch_aarch64 - Failed

vboxuser@UbuntuLTS:~/Desktop/zeek-6.0.0$ make -j$(nproc)
make -C build all
make[1]: Entering directory '/home/vboxuser/Desktop/zeek-6.0.0/build'
make[2]: Entering directory '/home/vboxuser/Desktop/zeek-6.0.0/build'
make[3]: Entering directory '/home/vboxuser/Desktop/zeek-6.0.0/build'
[ 0%] [BISON][BIFParser] Building parser with bison 3.8.2
[ 0%] [FLEX][BIFScanner] Building scanner with flex 2.6.4
make[3]: Leaving directory '/home/vboxuser/Desktop/zeek-6.0.0/build'
make[3]: Entering directory '/home/vboxuser/Desktop/zeek-6.0.0/build'
[ 0%] Building CXX object auxil/bifcl/CMakeFiles/bifcl.dir/bif_parse.cc.o
[ 0%] Building CXX object auxil/bifcl/CMakeFiles/bifcl.dir/bif_lex.cc.o
[ 0%] Building CXX object auxil/bifcl/CMakeFiles/bifcl.dir/bif_arg.cc.o
[ 0%] Building CXX object auxil/bifcl/CMakeFiles/bifcl.dir/module_util.cc.o
[ 0%] Linking CXX executable bifcl
make[3]: Leaving directory '/home/vboxuser/Desktop/zeek-6.0.0/build'
[ 0%] Built target bifcl
make[3]: Entering directory '/home/vboxuser/Desktop/zeek-6.0.0/build'
make[3]: Leaving directory '/home/vboxuser/Desktop/zeek-6.0.0/build'
make[3]: Entering directory '/home/vboxuser/Desktop/zeek-6.0.0/build'
[ 0%] [BIFCL] Processing /home/vboxuser/Desktop/zeek-6.0.0/auxil/zeek-af_packet-plugin/src/af_packet.bif

vboxuser@UbuntuLTS:~/Desktop/zeek-6.0.0$ sudo ps auxww | grep zeek
vboxuser  61213  0.0  0.0 17812 2244 pts/0    S+   17:20   0:00 grep --color=
auto zeek

vboxuser@UbuntuLTS:~/Desktop/zeek-6.0.0$ sudo zeekctl
vboxuser@UbuntuLTS:~/Desktop/zeek-6.0.0$ /usr/local/zeek/bin/zeek
^C
vboxuser@UbuntuLTS:~/Desktop/zeek-6.0.0$ /usr/local/zeek/bin/zeek --version
/usr/local/zeek/bin/zeek version 6.0.0

```

4)Brim

I downloaded the latest Zui .deb package

wget https://github.com/brimdata/zui/releases/download/v1.4.0/zui_1.4.0_amd64.deb

```

7715063848a0?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250503%2Fus-east-1%2F%2Faws
4_request&X-Amz-Date=20250503T143222Z&X-Amz-Expires=300&X-Amz-Signature=a04e8cac478c078f8503960db993dad522bfd18985893717
d813705ddddd5168&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dzui_1.4.0_amd64.deb&re
sponse-content-type=application%2Foctet-stream [following]
--2025-05-03 17:32:22-- https://objects.githubusercontent.com/github-production-release-asset-2e65be/144212075/1eb019f9
-6d19-4afc-a5f9-7715063848a0?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250503%2Fus-ea
st-1%2F%2Faws4_request&X-Amz-Date=20250503T143222Z&X-Amz-Expires=300&X-Amz-Signature=a04e8cac478c078f8503960db993dad52
2bfd18985893717d813705ddddd5168&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dzui_1.4
.0_amd64.deb&response-content-type=application%2Foctet-stream
Resolving objects.githubusercontent.com (objects.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110
.133, ...
Connecting to objects.githubusercontent.com (objects.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 139280246 (133M) [application/octet-stream]
Saving to: 'zui_1.4.0_amd64.deb'

  zui_1.  0%    0  --KB/s      zui_1.4  0% 335.33K  1.62MB/s      zui_1.4.  0% 687.33K  1.64MB/s
    zui_1.4.0  0%  1.01M  1.65MB/s      zui_1.4.0_  1%  1.36M  1.66MB/s      zui_1.4.0_a  1%  1.70M  1.6
5MB/s      zui_1.4.0_am  1%  2.05M  1.67MB/s      ui_1.4.0_amd  1%  2.39M  1.67MB/s      i_1.4.0_amd6  2%  2.73
M  1.67MB/s      _1.4.0_amd64  2%  3.08M  1.68MB/s      1.4.0_amd64.  2%  3.42M  1.68MB/s      .4.0_amd64.d  2%
3.76M  1.69MB/s      4.0_amd64.de  3%  4.11M  1.69MB/s      .0_amd64.deb  3%  4.45M  1.69MB/s      0_amd64.deb
3%  4.80M  1.69MB/s      _amd64.deb  3%  5.14M  1.69MB/s      etaamd64.deb  4%  5.48M  1.69MB/s      etamd64.d
eb  4%  5.83M  1.70MB/s      etad64.deb  64.deb  4.deb  4.deb  eb  b
zui_1.4.0_amd 100% 132.83M  649KB/s  in 59s

2025-05-03 17:33:22 (2.23 MB/s) - 'zui_1.4.0_amd64.deb' saved [139280246/139280246]

```

I installed the .deb package

```
sudo apt install ./zui_1.4.0_amd64.deb -y
```

```

vboxuser@UbuntuLTS:~/Desktop/zeek-6.0.0$ sudo apt install ./zui_1.4.0_amd64.deb -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'zui' instead of './zui_1.4.0_amd64.deb'
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-57 linux-headers-6.8.0-57-generic
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  zui
0 upgraded, 1 newly installed, 0 to remove and 1 not upgraded.
Need to get 0 B/139 MB of archives.
After this operation, 566 MB of additional disk space will be used.
Get:1 /home/vboxuser/Desktop/zeek-6.0.0/zui_1.4.0_amd64.deb zui amd64 1.4.0 [139 MB]
Selecting previously unselected package zui.
(Reading database ... 267184 files and directories currently installed.)
Preparing to unpack .../zeek-6.0.0/zui_1.4.0_amd64.deb ...
Unpacking zui (1.4.0) ...
Setting up zui (1.4.0) ...
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
N: Download is performed unsandboxed as root as file '/home/vboxuser/Desktop/zeek-6.0.0/zui_1.4.0_amd64.deb' couldn't be
accessed by user '_apt'. - pkgAcquire::Run (13: Permission denied)

```

```
vboxuser@UbuntuLTS:~/Desktop/zeek-6.0.0$ zui
17:39:13.049 > no brim data to migrate
17:39:13.061 > booting main with: {
  lakePort: 9867,
  lakeRoot: '/home/vboxuser/.config/Zui/lake',
  lakeLogs: '/home/vboxuser/.config/Zui/logs',
  lake: true,
  devtools: false,
  appState: '/home/vboxuser/.config/Zui/appState.json',
  autosave: true,
  releaseNotes: true,
  autoUpdater: true,
  singleInstance: true,
  pluginsPath: '/opt/Zui/plugins'
}
17:39:13.342 > initializers loaded
libva error: vaGetDriverNames() failed with unknown libva error
[68003:0503/173913.497100:ERROR:gpu_memory_buffer_support_x11.cc(44)] dri3 extension not supported.
```

